Multimodal Semantic Consistency-Based Fusion Architecture Search for Land Cover Classification

Xiao Li, *Student Member, IEEE*, Lin Lei, Caiguang Zhang, and Gangyao Kuang, *Senior Member, IEEE*

***Abstract*— Multimodal land cover classification (MLCC) using the optical and synthetic aperture radar (SAR) modalities has resulted in outstanding performances over using only unimodal data due to their complementary information on land properties. Previous multimodal deep learning (MDL) methods have relied on handcrafted multibranch convolutional neural networks (CNN) to extract the features of different modalities and merged them for land cover classification. However, natural-image-oriented handcrafted CNN models may not be the optimal strategies to handle remote sensing (RS) image interpretation problems, due to the huge difference in terms of imaging angles and imaging ways. Furthermore, few MDL methods have analyzed optimal combinations of hierarchical features from different modalities. In this article, we propose an efficient multimodal architecture search framework, namely, multimodal semantic consistency-based fusion architecture search (M^2SC-FAS) in continuous search space with the gradient-based optimization method, which can not only discover optimal optical- and SAR-specific architectures according to the different characteristics of the optical and SAR images, respectively, but also realizes the search of optimal multimodal dense fusion architecture. Specifically, the semantic consistency constraint is introduced to guarantee dense fusion between hierarchical optical and SAR features with high semantic consistency and then capture the complementary performance on land properties. Finally, the basis of curriculum learning strategy is adopted on M^2SC-FAS. Extensive experiments show superior performances of our work on three broad coregistered optical and SAR datasets.**

***Index Terms*— Convolutional neural network (CNN), heterogeneous semantic consistency, multimodal fusion, multimodal land cover classification (MLCC), network architecture search (NAS).**

I. Introduction

EARTH observation (EO) technologies have developed rapidly, and a massive number of multimodal remote sensing (RS) images have become more accessible in the same areas, which provides unprecedented opportunities for multimodal land cover classification (MLCC) [1], [2]. It aims to assign each unlabeled pixel in whole images with a specific label (e.g., tree, building and road) using the complementary contents between different modalities, which has been widely used in environmental monitoring [3], residential planning [4], and land resource management [5]. However, due to different imaging principles of optical and synthetic aperture radar (SAR) modalities, same land types from same modality data may display dissimilarities, and different land types from different modalities share more semantic similarities, which make the MLCC to be a considerably challenging task.

Great progresses have been made on multimodal deep learning (MDL) methods for image classification [6], object detection [7], and semantic segmentation [8] using multimodal data. Various MDL methods have also been widely used in the MLCC tasks due to their potent feature extraction capabilities [2], [9]. Generally, the MDL methods usually are composed of multiple modality-specific subnetworks that extract the discriminative unimodal features, and then project the multimodal features at a share space and merge them for computer vision tasks. Learning efficient aggregation feature representations is a pivotal problem in the MDL methods. According to the level of fusion, fusion strategies in the MDL methods can be achieved at early fusion, intermediately fusion, and decision fusion [10]. A massive number of studies [2], [4], [11] in the neuroscience [12] and machine learning [13] suggested the benefit of mid-level feature fusion. Due to the heterogeneous gaps in the optical and SAR RS images, it is not wise to use early fusion between optical and SAR data, as well as the decision fusion might lead to loss of lots of useful information and then reduce the performance of multimodal collaborative land cover classification [5], [6]. Therefore, our work mainly focuses on efficient mid-level feature fusion for improving the classification performance. Tremendous efforts [14], [15] on linear fusion paradigms have been studied to fuse discriminative features from different modality streams, and their simplicity and efficiency have made them widely used in multimodal computer vision [16], [17]. Another popular approach for mid-level fusion is bilinear pooling fusion [18], [19], which could obtain complicated relationships between different modalities features. MBFNet [1] is an improvement of the bilinear fusion model [18], which has introduced the second-order attention-based channel selection module to obtain compact multimodal bilinear fusion features for heterogeneous land cover classification. Following FBC [20], LC-BNet [21] introduced the locality attributes among similar samples to code sparse bilinear fusion features between different modalities for land cover classification with optical and SAR images.

Manuscript received 19 April 2022; revised 15 June 2022; accepted 14 July 2022. Date of publication 21 July 2022; date of current version 4 August 2022. *(Corresponding author: Lin Lei.)*

The authors are with the State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, National University of Defense Technology, Changsha 410073, China (e-mail: lixiao18@nudt.edu.cn; alaleilin@163.com; zhangcaiguang17@nudt.edu.cn; kuangyeats@hotmail.com).

Digital Object Identifier 10.1109/TGRS.2022.3193273

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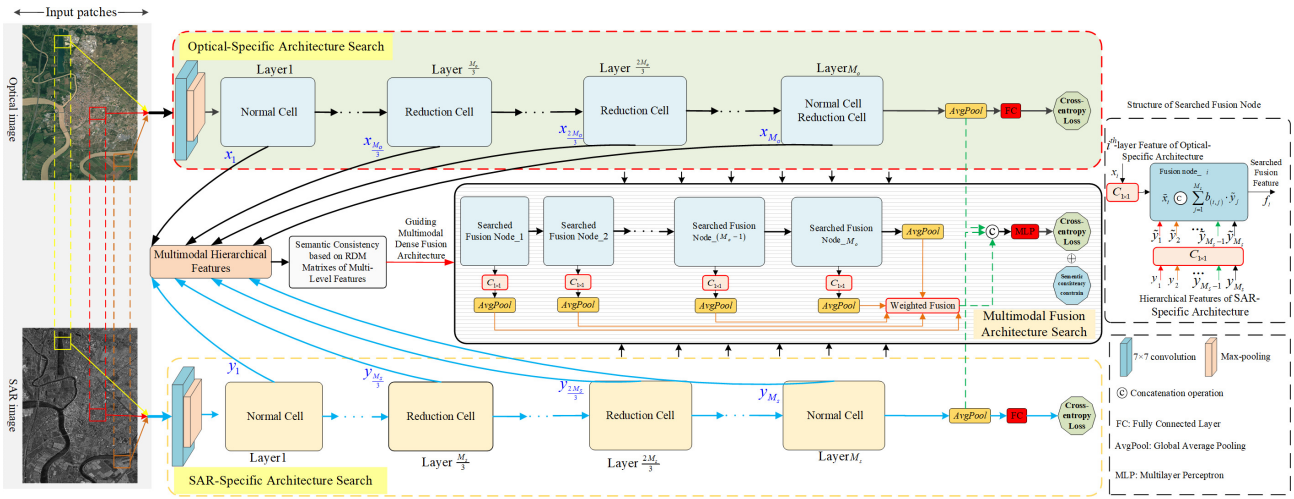


Fig. 1. Overall framework of our method for MLCC. First, a batch of raw optical and SAR images are used as the inputs of automatically designed optical- and SAR-specific CNN models, respectively, which contain multiple normal cells and reduction cells. Then, the parameters of optical- and SAR-specific CNN models are frozen, and then the optimal MSC-based dense fusion architecture is searched.

Despite promising performance, the major issues of the existing MDL methods with intermediate fusion mainly contain two aspects. On one hand, because the close-range images in the ImageNet dataset and the RS images are very different in terms of imaging angles and imaging methods, various handcrafted backbones are used in the MDL methods that may not be the optimal solution in the RS image interpretation problems [23], Many researchers [24], [25] have demonstrated that pretrained handcraft-based RS interpretation methods have been difficult to further improve due to the limitations of the manually designed network architecture, which proposed a series of convolutional neural architecture search (NAS) frameworks to automatically search the network structure for RS images and obtain better performance than the pretrained handcraft-based methods. On the other hand, in the traditional MDL methods, hierarchical features of the optical and SAR data consist of different semantic meanings, which are benefits for performance improvement. The existing MDL methods empirically believe that the corresponding mid-level features from different modality streams share a common space, which is beneficial to feature fusion. However, due to the heterogeneity of the optical and SAR data, the corresponding mid-level optical and SAR features do not share a common space, and the feature fusion strategies in the existing MDL methods are not most optimal fusion. Therefore, how to automatically obtain modality-specific feature extraction networks for optical and SAR data, respectively, and exploring what is the most optimal fusion strategy of hierarchical optical and SAR features for multimodal classification are critical.

NAS [26]–[30], termed automatic neural network search, aims to replace human experts with automatic models and design powerful task-special neural network architectures. The earliest researches on the NAS methods come from the literature [26], which used reinforcement learning with a meta-controller recurrent neural network (RNN) to select basic operating units (e.g., the number of filters, the filter sizes, and the types of activation functions), and then exploring a search space and obtain the optimal network, but requiring significant computational resources with 500 P100 GPUs over four days. Since then, to refine the search spaces, the evolutionary algorithm based on NAS has been proposed [27], [31]; however, these approaches still have a lot of computational complexity. More recently, advances [28], [32], [33] are applied to a one-shot setting of NAS, which search optimal cells and repeat searched cells to form whole network architecture. However, the above methods realized the architecture search on a discrete space, differentiable architecture search (DARTS) [29] optimized the architecture with fewer computational resources, which treated architecture search as black-box optimization problem in a continuous space. Most recently, the NAS methods have impressive results for multimodal learning tasks [26], [34]–[36], such as visual question answering (VQA) [36] and semantic segmentation [37].

Recently, NAS strategies have been investigated in the intelligent interpretation of the RS images [23]–[25], [38], [39], which aims to apply the characteristics of the RS data to guide the computer to automatically learn the network structure oriented to the RS domain, and then solve the limitations of the manually designed network architecture oriented to the natural images. Liu *et al.* [24] proposed a continuous particle-swarm-optimization-based deep learning architecture search model, which automatically designs cell-based convolutional neural network (CNN) architectures and obtains better performance over latest hyperspectral images (HIS) classification methods based on handcrafted CNN. Due to the enormous difference between natural images and RS images, it is inappropriate to directly migrate the natural-image-oriented existing CNN models for RS images' processing, Zhang *et al.* [25] proposed a more efficient search framework, RS-DARTS, to find the scene classification-oriented network architecture. Furthermore, to balance computational complexity and performance error of searched network, a framework for scene classification network architecture search based on multiobjective neural evolution (SceneNet) was proposed [38]. Peng *et al.* [23] proposed an efficient convolutional NAS to automatically search the RS-image-oriented network structure and obtained

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better performance than the pretrained handcraft-based methods. However, the above methods are only oriented to the network structure search in single-source RS tasks, and the multimodal dense fusion architecture search is rarely used in the field of multimodal RS images' interpretation, which can construct the optimal fusion strategy of multimodal RS data and efficiently delve the complementary information.

In this article, we propose a novel multimodal semantic consistency-based fusion architecture search (M^{2}SC-FAS) for MLCC using optical and SAR images, which contains both the design of modality-specific architecture for optical and SAR modalities and the dense-fusion-level design of multimodal features from modality-specific streams. Unlike previous works [2], [6] that explored same handcrafted backbones as feature extractor for each modality, we search the optical- and SAR-specific architectures according to the different characteristics of the optical and SAR RS data, to obtain excellent hierarchical unimodal features, respectively. Then, to obtain optimal hierarchical fusion features of different modality-specific streams, an NAS with semantic consistency based on representational dissimilarity matrix (RDM) [40] is introduced. Finally, the perspective of curriculum learning is used as the training paradigm of our method. We perform the searching procedure of our work to decide on the optimal multimodal dense fusion architecture and indicate the fruitfulness of our method on three broad coregistered datasets.

The main contributions can be summarized as:

1) We proposed a novel M^{2}SC-FAS framework, which automatically designs suitable multimodal dense fusion architecture in continuous search space with the gradient-based optimization method. The core idea is to measure semantic correlation of hierarchical optical and SAR features by introducing semantic consistency constraint, which ensures dense fusion of hierarchical optical and SAR features with high semantic consistency and then captures complementary information on land properties.
2) We discover the optical- and SAR-specific architectures according to the different characteristics of the optical and SAR data, respectively, which can alleviate the low discriminative features extracted by two modality-specific feature extractors with the same structure in the existing methods, due to heterogeneous gaps between optical and SAR data in the existing methods.
3) To the best of our knowledge, this article is the first to introduce an optimal multimodal dense fusion architecture between hierarchical optical and SAR features based on the NAS technology. We prove the fruitfulness of the proposed method on representative coregistered datasets. Comprehensive experiments present the searched modality-specific and multimodal dense fusion architectures achieving better classification results than the handcrafted-based CNN models.

II. Methodology

Fig. 1 shows the overall architecture of M^{2}SC-FAS. Step 1: We first discover optical- and SAR-specific architecture search spaces based on the DARTS algorithm, respectively.

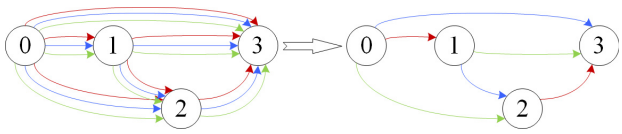


Fig. 2. DAG with an ordered sequence of intermediate nodes. Different colors on the edges denote different candidate operations.

Step 2: To search promising dense fusion architectures, mid-level features from different modalities are selected to fuse during search, and their associated semantic consistency constraint will be assigned to the objective. Step 3: The perspective of curriculum learning is used as the training paradigm of our method.

A. Modality-Specific Architecture Search Space

It is advisable to search the optical- and SAR-specific architectures due to their different imaging principles, respectively. The processes of architecture search are similar, and therefore, we uniformly introduce the architecture search of different modalities streams. A cell-based modality-specific architecture search contains the search of normal and reduction cells, and then they are repeatedly constructed to construct the final architecture. As a core searchable unit, a cell is defined as a directed acyclic graph (DAG) with an ordered sequence of intermediate nodes, as shown in Fig. 2. We suppose $s^{(j)}$ denotes the output of the jth intermediate node, which is calculated based on all its previous nodes

$$s^{(j)} = \sum_{i<j} o^{(i,j)}\left(s^{(i)}\right) \tag{1}$$

where $o^{(i,j)}(\cdot)$ denotes an operation (e.g., skip connection) in directed edge $(i; j)$, and $s^{(i)}$ denotes the outputs of the ith node before the jth node.

To make search space continuous, we set a parameter $a_o^{(i,j)}$ for the candidate operator o, and the outputs from candidate operations perform weighted summing to obtain continuous representation of the mixture layer. The architecture weight w of all operations space $\mathcal{O}$ is computed based on the soft-max function, as following:

$$w_o^{(i,j)} = \frac{\exp\left(a_o^{(i,j)}\right)}{\sum_{o'\in\mathcal{O}} \exp\left(a_{o'}^{(i,j)}\right)} \tag{2}$$

and the operation for each edge is parameterized as a soft-max mixture over all the possible operations

$$\bar{o}^{(i,j)}(s) = \sum_{o\in\mathcal{O}} w_o^{(i,j)} o(s) \tag{3}$$

where $a_o^{(i,j)} \in {}^{N_o}$ denotes the scores of all candidate operations on edges from $s^{(i)}$ to $s^{(j)}$, which is a learnable parameter of the operation $o(\cdot)$. Then, we focus on learn continuous variables $\alpha = \{\alpha^{(i,j)}\}$ and the weights w of network architecture over all mixed operations, and then obtain the optimal network architecture.

The optimization of α and w is referred to as a bilevel optimization problem. The search of architecture is to find α^*

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by minimizing validation loss $\mathcal{L}_{\text{val}}(w^*, \alpha)$, while the learning of architecture weight is solved by minimizing training loss $\mathcal{L}_{\text{train}}(w, \alpha^*)$

$$\begin{aligned} &\min_{\alpha} \mathcal{L}_{\text{val}}\big(w^*(\alpha), \alpha\big) \\ &\text{s.t.} \quad w^*(\alpha) = \arg\min_{w} \mathcal{L}_{\text{train}}(w, \alpha). \end{aligned} \tag{4}$$

In our work, $\mathcal{O}$ consists of three types of operations, i.e., the shortcut, max and average pooling and identity, the learned operations, separable convolution, and dilated convolution. Separable convolution can efficiently balance computational burden and experimental performances. The operator space $\mathcal{O}$ is collected specifically as:

Then, we obtain optimal optical- and SAR-specific architectures based on (4), respectively.The optimal numbers of cells $\{M_o, M_s\}$ and nodes $\{N_o, N_s\}$ in the optical- and SAR-specific architectures are presented in Fig. 4.

B. Multimodal Dense Fusion Architecture Search Space

Based on the searched optical and SAR streams, we have the hierarchical optical features $\{x_i\}_{i=1}^{M_0}$ and the SAR features $\{y_j\}_{j=1}^{M_s}$ from different cells. We define the multimodal fusion search space as the arrangement of cells from different modality-specific streams, which controls the ways of mix features with varying levels at optical and SAR modalities. We associate a scalar $b_{(i,j)}$ with the fusion between the feature x_i from the ith cell of the optical stream and the feature y_j from the jth cell of the SAR stream, where $b_{(i,j)}$ is the element in the ith row and the jth column of multimodal search space $\mathcal{B}$. Taking the ith fusion node as a reference, the candidate hierarchical SAR feature $\{y_j\}_{j=1}^{M_s}$ performs weighted sum based on a scalar $b_{(i,j)}$, and then the output of the ith fusion node is obtained by concatenating optical feature $\tilde{x}_i$ and weighted feature $\sum_{j=1}^{M_s} b_{(i,j)} \cdot \tilde{y}_j$, as follows:

$$\begin{cases} \tilde{x}_i = \text{Conv}(x_i) \\ \tilde{y}_j = \text{Conv}\big(\text{Samp}\big(y_j\big)\big) \\ f_i = \tilde{x}_i © \sum_{j=1}^{M_s} b_{(i,j)} \cdot \tilde{y}_j \end{cases} \tag{5}$$

where Samp$(\cdot)$ denotes the linear interpolation operation, which is used to sample y_j to be the same size as x_i, and Conv$(\cdot)$ denotes the 1×1 convolution with 128 kernels that is used in uniform arbitrary feature dimensions, and © denotes the features' concatenation operation in the channel dimension.

Then, the output feature f_i of the ith fusion node is input into the convolutional layer so as to learn the correlations between different modalities' features. We perform global average pooling AvgPool$(\cdot)$ to uniform feature dimension and then obtain fusion feature vector $\tilde{f}_i$ from the ith fusion node. This is done according to the following equation:

$$\tilde{f}_i = \text{AvgPool}\big(\text{Conv}(f)_i\big). \tag{6}$$

Similarly, we can obtain the fusion feature vectors $\{\tilde{f}_i\}_{i=1}^{M_o}$ of all the fusion nodes $(\{i\text{th}\}_{i=1}^{M_o})$. We perform the weighted fusion highlighting the varying contributions of different-level fusion features

$$\tilde{f} = \widehat{w}_1 \tilde{f}_1 © \cdots © \widehat{w}_{M_o} \tilde{f}_{M_o} \tag{7}$$

where $\widehat{w} = \{\widehat{w}_1, \ldots, \widehat{w}_{M_o}\}$ denotes the learnable weight vectors.

Finally, we connect weighted fusion feature $\tilde{f}$, optical feature x_N, and SAR feature y_M that are input to the multilayer perceptron (MLP) with a hidden layer for land cover classification. This is done according to the following equation:

$$\bar{f} = \text{Mlp}\big(\tilde{f} © x_{M_o} © y_{M_s}\big). \tag{8}$$

To make multimodal fusion search space continuous, referring to the operation for $a_o^{(i,j)}$ in (2) and (3), we relax the categorical choice of a particular operation $b_{(i,j)}$ to a softmax over all possible operations $\mathcal{B}$.

Especially, to guide the learning of multimodal dense fusion architecture search space $\mathcal{B}$, and then obtain the optimal dense fusion way of hierarchical features from the optical and SAR data, the semantic consistency constraint based on the RDM is introduced into the objective function. For a given feature matrix Z, we obtain the RDM (M^Z) by calculating the pairwise distances between each pair of features vectors based on the correlation residual

$$M^Z_{(i,j)} = 1 - \text{corr}\big(z_i, z_j\big) \tag{9}$$

where corr$(\cdot)$ denotes the correlation calculation operation based on the cosine similarity, and z_i and z_j denote the rows i and j in the matrix Z.

For $\{x_i\}_{i=1}^{M_o}$ and $\{y_j\}_{j=1}^{M_s}$, we derive the semantic consistency matrix V using the correlation between RDMs of the optical and SAR features (i.e., the correlation $v_{(i,j)}$ between M^{x_i} and M^{x_j} measures the semantic consistency between x_i and y_j)

$$V \in \Re^{M_o \times M_s}, v_{(i,j)} = \text{corr}\big(M^{x_i}, M^{x_j}\big) \tag{10}$$

where $v_{(i,j)}$ is the element of the ith row and jth column of V.

Furthermore, V is used to construct a semantic consistency constraint $\tilde{\mathcal{L}}_{sc} = \sum_{i=1}^{M_o} \sum_{j=1}^{M_s} v_{(i,j)} \cdot b_{(i,j)}$, where $b_{(i,j)}$ is the element in the ith row and jth column of $\mathcal{B}$. The semantic consistency constraint guides the learning of $\mathcal{B}$ and then obtains the optimal way of mix features with varying levels at different modalities.

The multimodal dense fusion architecture search problem is reduced to learn fusion architecture $\mathcal{B}^*$ by minimizing the joint loss using validation set, while the architecture weight problem is reduced to learn network parameter $w_f{}^*$ by optimizing the training loss using the training set, as follows:

$$\begin{aligned} &\min_{w_f} \tilde{\mathcal{L}}_{\text{val}}\big(\mathcal{B}^*\big(w_f\big), w_f\big) + \lambda \tilde{\mathcal{L}}_{sc} \\ &\text{s.t.} \quad \mathcal{B}^*\big(w_f\big) = \arg\min_{\mathcal{B}} \tilde{\mathcal{L}}_{\text{train}}\big(\mathcal{B}, w_f\big) \end{aligned} \tag{11}$$

where λ is a balance coefficient, and w_f consists of the network parameters w_o and w_s of the optical-specific and SAR-specific streams that are frozen, and the network parameters of dense fusion architecture that need to be trained.

After the architecture parameter $\mathcal{B}$ is obtained, the final discrete dense fusion architecture is derived by: $b_{(i,j)} = \arg\max_j b(i,:)$, where $\tilde{x}_j$ and $\tilde{y}_j$ are the optimal combination.

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C. Training Paradigm

We think about the training paradigm of our method from the perspective of curriculum learning, which mainly consists of two steps: modality-specific architecture search and multimodal dense fusion architecture search. The detailed training steps are: First, we use the search algorithm to search the optimal optical- and SAR-specific architectures based on (4), respectively. After the optimal optical and SAR architectures are searched, we search the optimal multimodal dense fusion architecture based on (11), where the parameters of the searched optical- and SAR-specific streams are frozen. The training process of the proposed M^{2}SC-FAS is shown in Algorithm 1.

Algorithm 1 Framework of the proposed M^{2}SC-FAS method

Initial modality-specific architecture $\tilde{\alpha}_o^{(i,j)}$ and $\bar{\alpha}_o^{(i,j)}(o \in \mathcal{O})$ for each edge (i, j) in a cell, and initial multimodal dense fusion architecture $\mathcal{B}$, the number of cells $\{M_o, M_s\}$ and nodes $\{N_o, N_s\}$ for optical- and SAR-specific network architectures, the number of initial channels $C = 32$.

1. Optical architecture search:

1.1: Initialize the optical network weight w_o, learning rate ξ_0, and ε_0;

1.2: **while** not converged **do**

1.3 Update the optical architecture $\tilde{\alpha}_o$ based on (4)

$\tilde{\alpha}_o = \tilde{\alpha}_o - \xi_o \nabla_{\tilde{\alpha}_o} \mathcal{L}_{val}(w_o - \xi_o \nabla_{w_o} \mathcal{L}_{train}(w_o, \tilde{\alpha}_o), \tilde{\alpha}_o)$;

1.4: Update the optical network weights w_o

$w_o = w_o - \varepsilon_o \nabla_{\tilde{\alpha}_o} \mathcal{L}_{train}(w_o, \tilde{\alpha}_o)$;

2. SAR architecture search:

2.1: Initialize the SAR network weights w_s, learning rate ξ_s, and ε_s;

2.2: **while** not converged **do**

2.3: Update the SAR architecture $\bar{\alpha}_o$ based on (4)

$\bar{\alpha}_o = \bar{\alpha}_o - \xi_s \nabla_{\bar{\alpha}_o} \mathcal{L}_{val}(w_s - \xi_s \nabla_{w_s} \mathcal{L}_{train}(w_s, \bar{\alpha}_o), \bar{\alpha}_o)$;

2.4: Update the SAR network weights w_s

$w_s = w_s - \varepsilon_s \nabla_{\bar{\alpha}_s} \mathcal{L}_{train}(w_s, \bar{\alpha}_o)$;

3. Dense fusion architecture search:

3.1: Derive the final two optical- and SAR-specific architectures based on architecture $\tilde{\alpha}_o^{(i,j)}$ and $\bar{\alpha}_o^{(i,j)}$;

3.2: Initialize dense fusion network weights $\tilde{w}$ based on w_o and w_s, learning rate ξ_f, and ε_f;

3.3: **while** not converged **do**

3.4 Update the fusion architecture $\mathcal{B}$ based on (11)

$\mathcal{B} = \mathcal{B} - \xi_f \nabla_{\mathcal{B}} \mathcal{L}_{val}(w_f - \xi_f \nabla_{w_f} \mathcal{L}_{train}(w_s, \mathcal{B}), \mathcal{B})$;

3.5: Update the fusion network weights w_f

$w_f = w_f - \varepsilon_f \nabla_{w_f} \mathcal{L}_{train}(w_f, \mathcal{B})$;

3.6: Output fusion architecture $\mathcal{B}$ and derive final fusion architecture.

III. Experiments and Discussion

A. Experimental Setups

1) Datasets: We estimate the effectiveness of M^{2}SC-FAS on the Libourne, Bergerac, and PoDelta datasets, as shown in Fig. 3. Libourne comes from the region (distributed between the (44.53 °N, 0.10 °E) upper left coordinates and (44.57 °N, 0.16 °E) lower right coordinates) of Libourne city, Bordeaux, France, which includes GoogleEarth data and TerraSAR with Horizontal transmit-Horizontal receive (HH) polarization imaging mode data acquired on June 28 and September 22, 2009, respectively. Eventually, both GoogleEarth and TerraSAR images have been coregistered and the final size of 6384 × 3542 with 3.0 m × 3.0 m is obtained. The ground truth of Libourne is obtain based on Open Street Maps [37] and visual interpretation, which includes seven land types. Bergerac comes from the region (distributed between the (44.54 °N, 0.24 °E) upper left coordinates and (44.57 °N, 0.28 °E) lower right coordinates) of Bergerac city, France, which includes GoogleEarth data and TerraSAR with HH-polar imaging mode data acquired on September 29 and 30, 2016, respectively. Eventually, both GoogleEarth and TerraSAR images have been coregistered and the final size of 4577 × 4181 with 3.0 m × 3.0 spatial resolution is obtained. The ground truth of Bergerac is obtained based on Open Street Maps [41] and visual interpretation, which includes eight land types. PoDelta comes from a region of PoDelta, Italy, which includes GoogleEarth data and COSMO-SkyMed with HH-polar imaging mode data, respectively. Both GoogleEarth and COSMO-SkyMed images have been coregistered and the final size of 3666 × 1752 with 3.0 m × 3.0 spatial resolution is obtained. We relabel the ground truth of PoDelta provided by the literature [1], [2], which includes five land types.

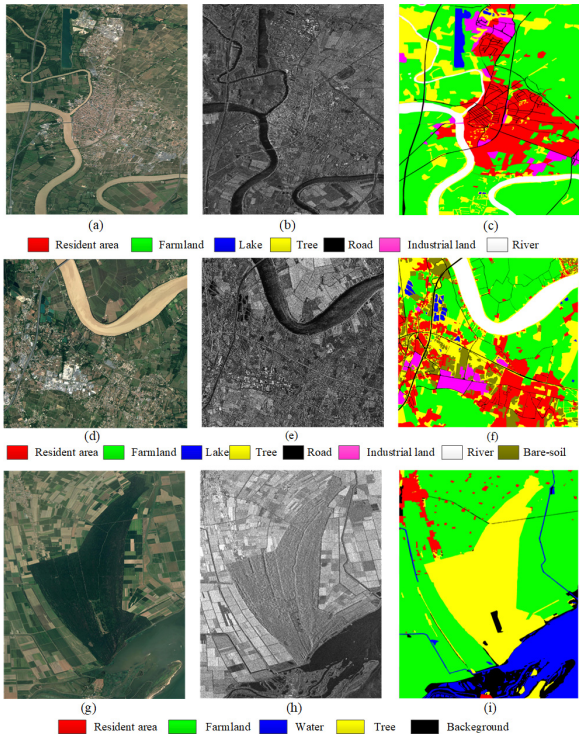


Fig. 3. Visualization of the Libourne, Bergerac, and PoDelta datasets. (a), (d), and (g) Optical images. (b), (e), and (h) SAR images. (c), (f), and (i) Ground truths. Libourne contains seven land cover categories: resident area, river, road, farmland, tree, industrial area, and lake. Bergerac contains eight land cover categories: resident area, farmland, lake, tree, road, industrial area, river, and bare soil. PoDelta contains five land cover categories: resident area, water, farmland, tree, and background.

2) Implementation Details: We first use the differentiable search algorithm to find optical- and SAR-specific architectures using the optical and SAR data, respectively. After the optimal unimodal architectures are obtained, the fusion architecture is searched through the gradient descent algorithm. In the modality-specific search phase, an Adam optimizer [46] with a base learning rate of $\xi = 1e{-}4$, a weight decay

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$\beta = 3e{-}8$, and a momentum $\eta = (0.98, 0.999)$ is used to learn architecture parameters $\tilde{\alpha}_o$ and $\bar{\alpha}_o$. We use the stochastic gradient descent (SGD) algorithm [36] to optimize the weights w_o of the optical stream and the weights w_s of the SAR streams with an initial learning $\varepsilon = 1e - 3$, a momentum 0.95, and a weight decay $\beta = 3e{-}8$. The batch size is 32, and the epoch is 50. After the modality-specific architectures are searched and used as the backbone to fusion networks, an Adam optimizer with a base learning rate of $\xi_f = 1e{-}5$, a weight decay $\beta = 3e{-}8$, and a momentum $\eta = (0.98, 0.999)$ is used to learn fusion architecture parameter $\mathcal{B}$. The SGD algorithm is used to optimize the fusion network weights w_f with an initial learning $\varepsilon_f = 1e{-}4$, a momentum 0.95, and a weight decay $\beta = 3e{-}8$. For the Libourne and Bergerac datasets, the optimal patch size is 35×35. For the PoDelta dataset, the optimal patch size is 29×29. The numbers of training and validation samples in the Libourne, Bergerac, and PoDelta datasets for each land type are 30 000 and 5000, respectively, and the remainders are used to test sets. We have performed fivefold cross validation experiments on three datasets to verify the effectiveness of the proposed method M^{2}SC-FAS.

B. Comparative Methods

In the section, we describe comparison methods in detail, which mainly concern three groups: unimodal search models, multimodal fusion methods, and multimodal fusion search models, as following:

Unimodal search methods: To verify our model can learn complementary information, we compare the proposed M^{2}SC-FAS with single-modal models.

1) SCNN [47]: Liu *et al.* used the statistical properties of an RS image to present a statistical CNN (SCNN) for land cover classification.

2) Suoyan *et.al* [49]: Pan *et al.* constructed a light CNN model to obtain high land cover classification performance.

3) Unimodal-NAS-Opt.: We search optimal optical-specific network architecture based on (4), and the optical modality is used as the input samples.

4) Unimodal-NAS-SAR: We search optimal SAR-specific network architecture based on (4), and the SAR modality is used as the input samples.

Multimodal fusion methods: To analyze the efficiency of the proposed semantic consistency-based fusion architecture search in our model, the multimodal fusion methods are described below:

1) CNN-Opt.&SAR: CNN-Opt.&SAR method comes from the literature [50], which is trained based on the optical and SAR modalities for land cover classification.

2) MultiCNN: Xu *et al.* [44] put forward a two-brand convolution network for HIS and Light Detection and Ranging (LiDAR) classification. To adapt our multimodal situation, a two-tunnel convolution subnetwork deals with the optical modality data, and the subnetwork with cascade block realizes feature extraction of the SAR modality data.

3) B-CNN: Lin *et al.* [18] used bilinear pooling on two feature maps from the optical and SAR streams and applied element-wise sign square root normalization and l_2 normalization prior to solve visual burstiness. Finally, a softmax classifier is used for complete classification.

4) FBC: Gao *et al.* [20] proposed factorized bilinear coding to obtain sparse bilinear features. To adapt our multimodal situation, FBC is used as an efficient fusion strategy for land cover classification, and the pretrained AlexNet is used as the backbone.

5) MLB: Kim *et al.* [45] put forward a multimodal low-rank bilinear attention network to obtain compact bilinear features for classification. To adapt our multimodal situation, we can obtain bilinear fusion features between the optical and SAR features using low-rank bilinear pooling.

6) LC-BNet [21]: Following FBC [20], LC-BNet introduced the locality attributes among similar samples to code sparse bilinear fusion features between different modalities.

7) CHGFNet [2]: Li *et al.* introduced the multimodal collaborative attention module to capture the correlation between different modalities features, and the gated heterogeneous fusion module learned varying contributions of the optical and SAR modalities.

8) MBFNet [1]: This model was an improvement of the bilinear fusion modal, which introduces the second-order attention-based channel selection module to obtain compact multimodal bilinear fusion features.

Multimodal fusion search models: To analyze the effectiveness of our model, we compare our model with the following multimodal fusion search models:

1) Random search: This method is a variant of our model, which directly obtains fusion architecture based on the random search strategy [36], instead of the semantic consistency-based fusion architecture search.

2) MFAS [51]: Prez-Ra *et al*. introduced a sequential model-based optimization [42] scheme to search optimal multimodal feature fusion, instead of semantic consistency-based fusion architecture search. The pretrained models were used as backbones of different modality-specific streams, instead of the searched architecture.

C. Hyperparameter Configuration

1) Effects of Numbers of Cells and Nodes: The parameters $\{M_o, M_s\}$ and $\{N_o, N_s\}$ determine the architectures of the optical- and SAR-specific streams, respectively. In Fig. 4, the x-axis denotes the numbers of cells $\{M_o, M_s\}$, the y-axis denotes the numbers of nodes $\{N_o, N_s\}$, and the z-axis denotes the PA of Unimodal-NAS-SAR and Unimodal-NAS-Opt. Taking Fig. 4(c1) and Fig. 4(c2) as an example, the classification curves have an increasing trend with the increase in the numbers of cells and nodes. When the numbers of cells and nodes are $\{9, 12\}$ and $\{4, 3\}$, respectively, the classification is optimal in PaDelta, followed by the classification accuracies with the increase in the numbers of cells and nodes, which means larger architectures lead to overfit. Furthermore, when the numbers of cells and nodes are $\{10, 12\}$ and $\{4, 3\}$, respectively, the classification performances M^{2}SC-FAS are optimal in the Libourne dataset. When the numbers of cells and nodes are $\{12, 13\}$ and $\{4, 3\}$, respectively, the classification

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TABLE I

LAND TYPES' DISTRIBUTION ON THE LIBOURNE, BERGERAC, AND PODELTA DATASETS, LAND COVER TYPES IN BRACKETS ARE THE PODELTA DATASET

#	Land Types	Libourne		Bergerac		PoDelta	
		Number	Percentage	Number	Percentage	Number	Percentage
1	Resident area	3999047	17.97%	359938	18.81%	213868	2.99%
2	Lake (Water)	355261	53.58%	210512	1.10%	826053	11.55%
3	Farmland	11890375	53.41%	7081777	37.01%	497418	6.95%
4	Tree	2949197	13.25%	3525715	18.42%	3834510	53.58%
5	Road (Background)	723753	3.25%	620455	3.24%	1784183	24.93%
6	Industrial land	914262	4.11%	674799	3.53%	–	–
7	River	1426033	6.41%	1845398	9.64%	–	–
8	Bare-soil	–	–	1578410	8.35%	–	–

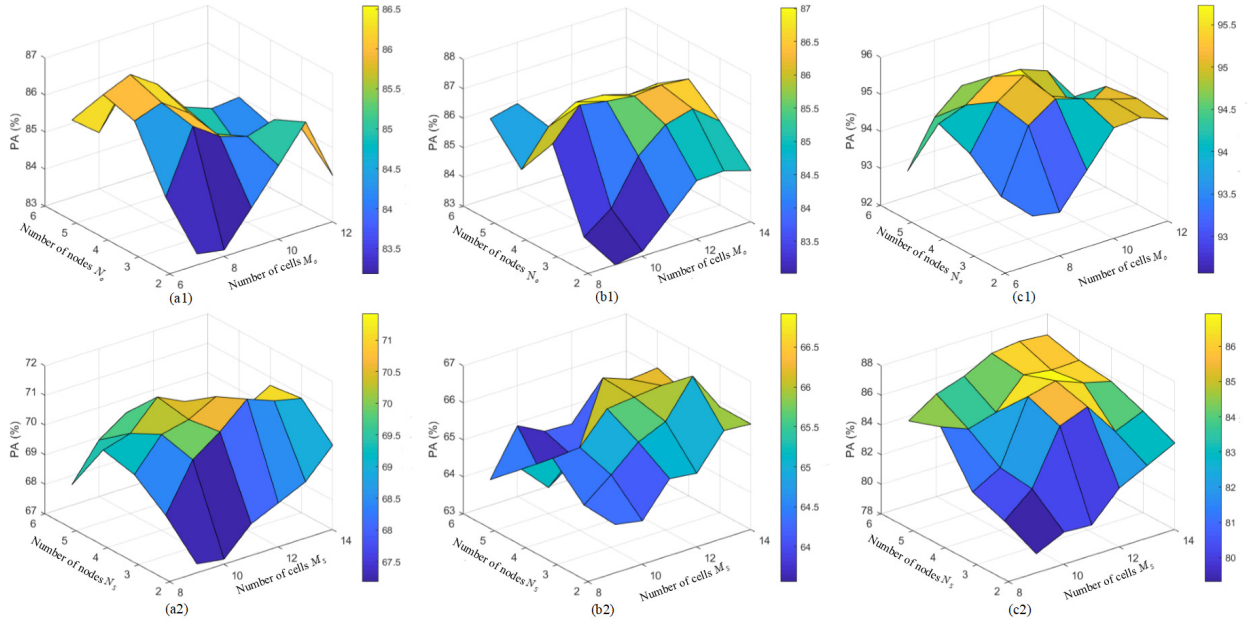


Fig. 4. Influence of the numbers of cells $\{M_o, M_s\}$ and nodes $\{N_o, N_s\}$ on the Libourne, Bergerac, and PoDelta datasets. (a1), (b1), and (c1) Trends between the accuracies and the numbers of cells M_o and the numbers of cells N_o for optical-specific stream on the Libourne, Bergerac, and PoDelta datasets. (a2), (b2), and (c2) Trends between the accuracies and the numbers of cells M_s and the numbers of cells N_s for SAR-specific stream on the Libourne, Bergerac, and PoDelta datasets.

performances are optimal in the Bergerac dataset. It is worth noting that the searched optical stream contains shallow layers, while the searched SAR stream contains deeper layers due to the weak explainability of the SAR images, and the optical stream contains wider cell architecture that enriches the diversity of features at the same level.

2) Effects of the Parameter λ: The parameter λ in (11) maintains the balance between the performance and multimodal semantic consistency (MSC). In Fig. 5, the horizontal axis $\{0, 1e{-}5, 1e{-}4, 1e{-}3, 1e{-}2, 1e{-}1, 1\}$ denotes λ, and the vertical axis denotes PA. Different values of λ guide the learning of different multimodal dense fusion architecture search spaces $\tilde{w}$. When $\lambda = 0$, MSC is lost in multimodal dense fusion architecture search, and the classification performances are not optimal in all the datasets, which can verify that the introduced MSC can guide the search of the optimal multimodal dense fusion architecture. When $\lambda = 1e{-}4$, $\lambda = 1e{-}5$, and $\lambda = 1e{-}3$, the M^{2}SC-FAS obtains optimal classification performances in the Libourne, Bergerac, and PoDelta datasets, respectively. The curves have a smoother change with the change in parameter λ, which means our method is not very sensitive to the changes in the parameters.

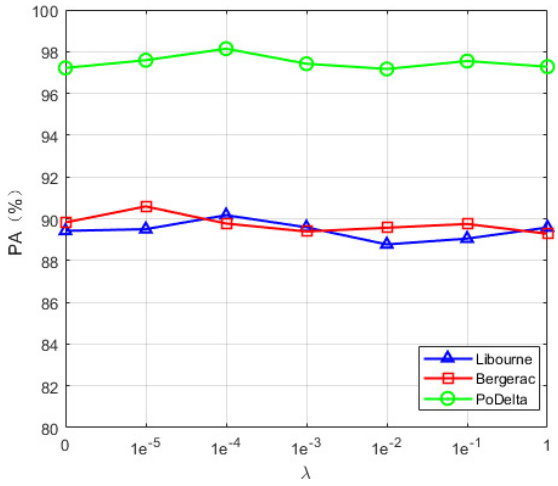


Fig. 5. Curves to depict the trends between the classification accuracies and parameter λ on the Libourne, Bergerac, and PoDelta datasets, respectively.

D. Analysis of Search Results

We analyze the complexity of our proposed framework. The inceptive search space contains eight candidate operations

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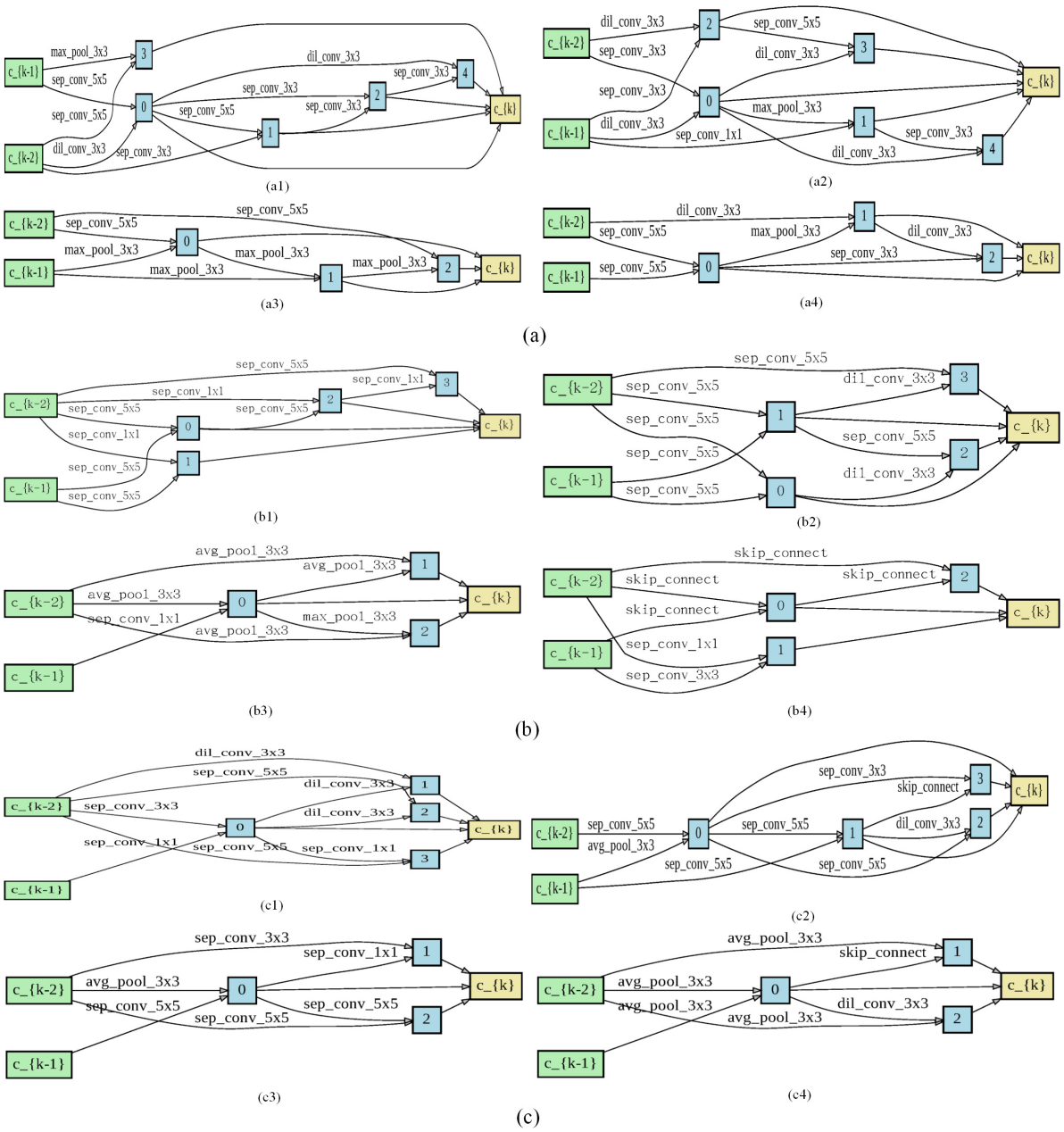


Fig. 6. Searched cell architectures of the optical- and SAR-specific streams on the Libourne, Bergerac, and PaDelta datasets, respectively. (a) Searched normal and reduction cells for optical- and SAR-specific architecture on the Libourne dataset. (b) Searched normal and reduction cells for optical- and SAR-specific architecture on the Bergerac dataset. (c) Searched normal and reduction cells for optical- and SAR-specific architecture on the PaDelta dataset. (a1), (b1), and (c1) Normal cell of optical-specific architecture. (a2), (b2), and (c2) Reduction cell of optical-specific architecture. (a3), (b3), and (c3) Normal cell of SAR-specific architecture. (a4), (b4), and (c4) Reduction cell of SAR-specific architecture.

$\mathcal{O}$ for modality-specific architecture search. We set different numbers of cells and nodes for different coregistered optical and SAR datasets, as shown in Fig. 4. Theoretically, each discretized cell allows $\varphi_o = \prod_{i=1}^{N_o} (i(i+1)/2) \times N_o^2$ and $\varphi_s = \prod_{i=1}^{N_s} (i(i+1)/2) \times N_s^2$ possible DAGs without graph isomorphism in optical and SAR streams, respectively. Since the normal and reduction cells are simultaneously trained during the search phase, the numbers of the optical and SAR architectures are almost φ_o^2 and φ_s^2, respectively. In multimodal dense fusion architecture search space, the number of fusion architectures depends on the number of cells $\{M_o, M_s\}$ from the optical and SAR streams, separately, which is calculated as $M_o M_s$. Therefore, the total number of our method is $(\varphi_o^2 + \varphi_s^2 + M_o M_s)$. The reduction cells are inserted at a network depth of 1/3 and 2/3 in the optical and SAR streams, respectively. Fig. 6 illustrates the searched cell architectures of the optical- and SAR-specific streams on the Libourne, Bergerac, and PaDelta datasets, respectively, and repeat searched cells to form the whole network architecture for the final MDL classification results.

In addition, visualization of multimodal fusion search spaces $\mathcal{B}$ of the proposed M^{2}SC-FAS on the Libourne, Bergerac, and

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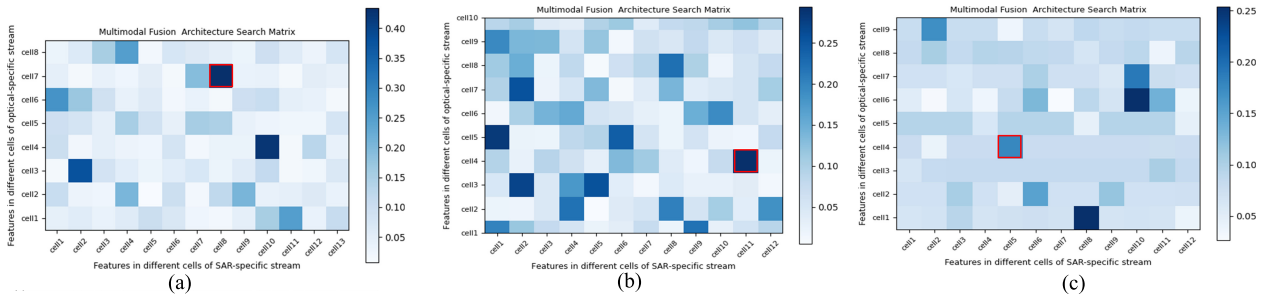


Fig. 7. Visualization of multimodal dense fusion search space matrixes $\mathcal{B}$ on the Libourne, Bergerac, and PaDelta datasets, respectively. The highlighted square in the spatial position (i, j) of fusion space matrixes $\mathcal{B}$ denotes the mixing way between the optical feature $\tilde{x}_i$ and the SAR feature $\tilde{y}_j$. (a) Multimodal fusion architecture search matrix on the Libourne dataset. (b) Multimodal fusion architecture search matrix on the Bergerac dataset. (c) Multimodal fusion architecture search matrix on the PaDelta dataset.

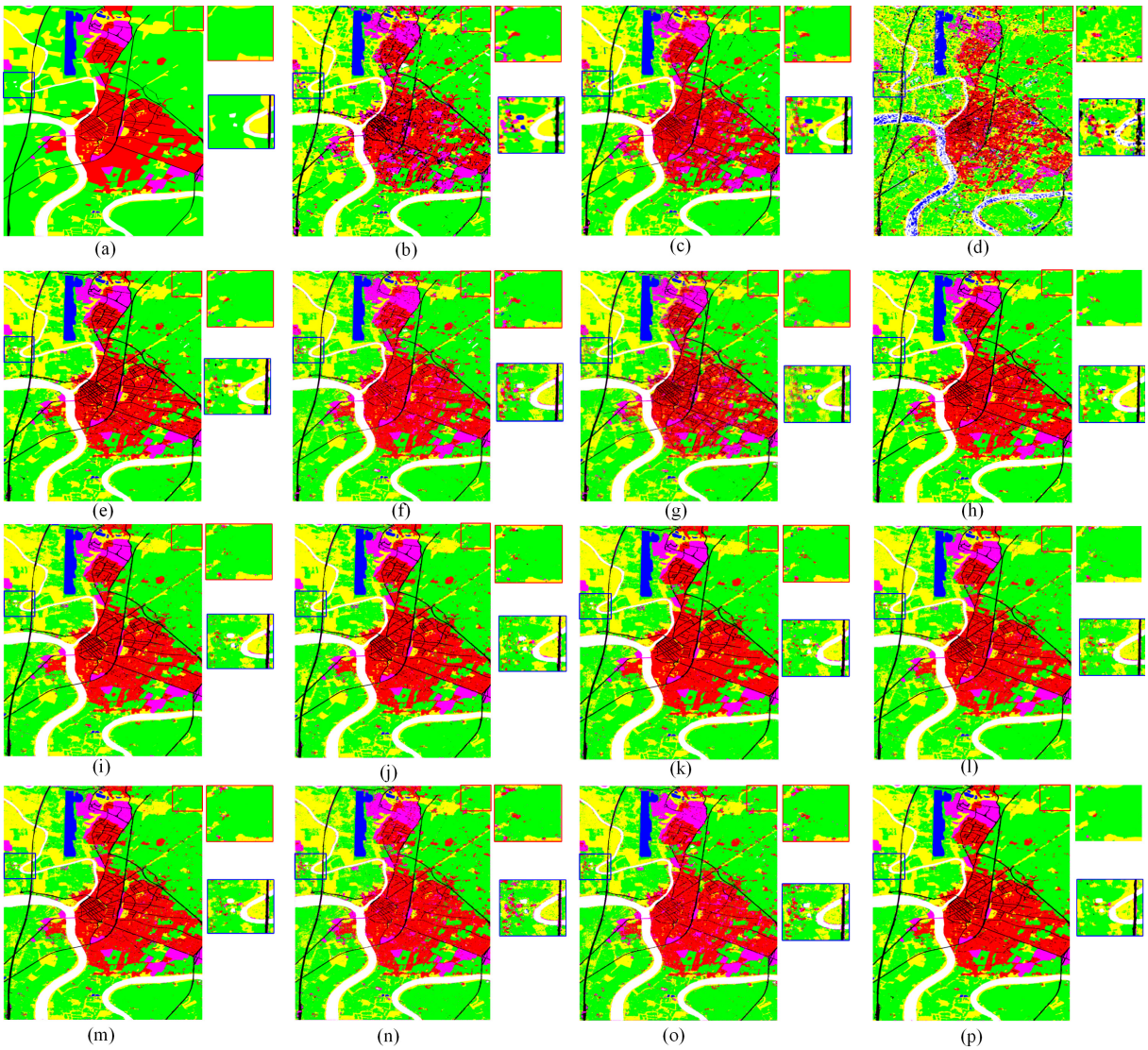


Fig. 8. Classification maps of different methods on Libourne. (a) Ground truth. (b) SCNN [47]. (c) Suoyan *et al.* [49]. (d) Unimodal-NAS-SAR. (e) Unimodal-NAS-Opt. (f) CNN-Opt.&SAR. (g) MultiCNN [44]. (h) B-CNN [18]. (i) FBC [20]. (j) MLB [45]. (k) LC-BNet [21]. (l) CHGFNet [2]. (m) MBFNet [1]. (n) Random search. (o) MFAS [51]. (p) M^{2}SC-FAS (our).

PaDelta datasets, respectively, as presented in Fig. 7. Taking Fig. 7(a) as an example, the highlighted red square in the spatial position (i, j) of fusion space matrixes $\mathcal{B}$ denotes the optimal mixing way between features $\tilde{x}_i$ and $\tilde{y}_j$, e.g., $b_{(7,8)}$ in Fig. 7(a) denotes the optical feature $\tilde{x}_7$ and the SAR feature $\tilde{y}_8$ should be fused.

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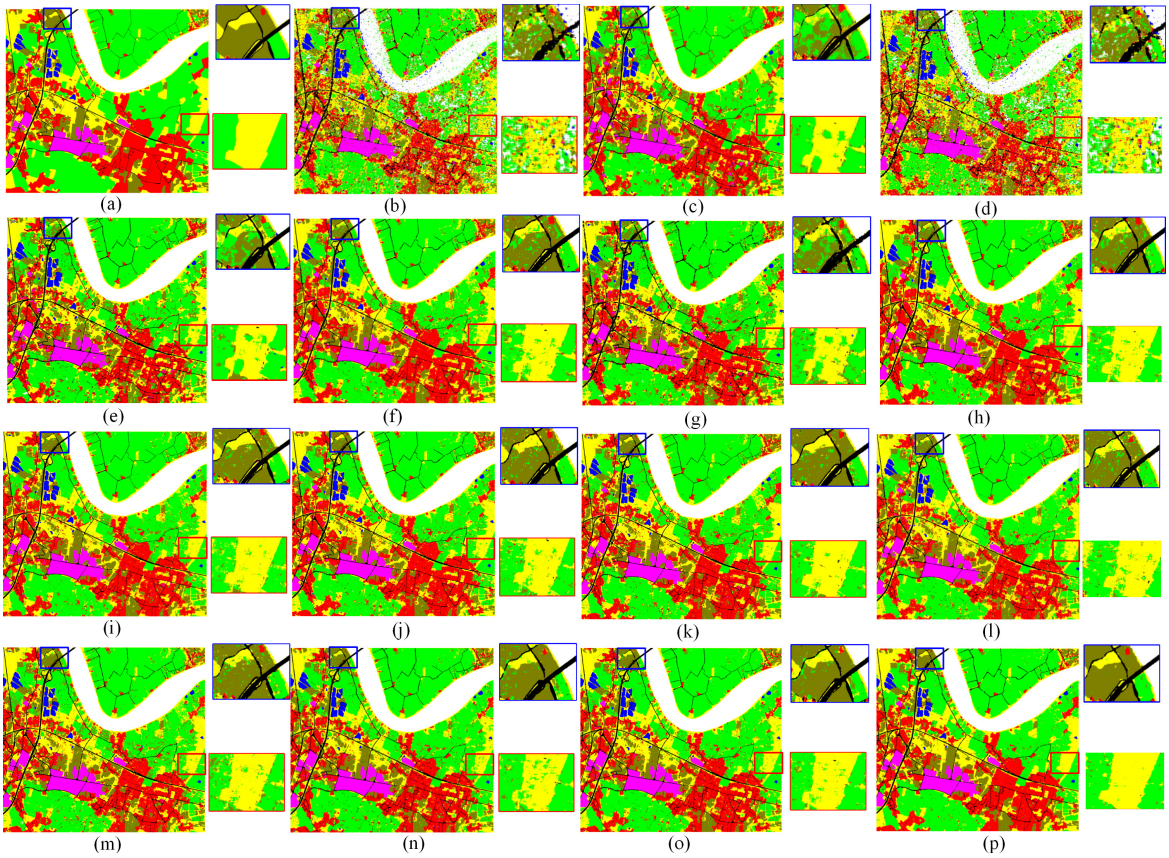


Fig. 9. Classification maps of different methods on Bergerac. (a) Ground truth. (b) SCNN [47]. (c) Suoyan *et al.* [49]. (d) Unimodal-NAS-SAR. (e) Unimodal-NAS-Opt. (f) CNN-Opt.&SAR. (g) MultiCNN [44]. (h) B-CNN [18]. (i) FBC [20]. (j) MLB [45]. (k) LC-BNet [21]. (l) CHGFNet [2]. (m) MBFNet [1]. (n) Random search. (o) MFAS [51]. (p) M^{2}SC-FAS (our).

E. Evaluation Results

We estimate the classification performance of our method on the Libourne, Bergerac, and PoDelta datasets, respectively. Table II indicates the land cover classification results of the proposed M^{2}SC-FAS and three categories of contrast methods: 1) Unimodal search methods: SCNN [47] (opt.) and Suoyan *et.al* (opt.) [49], which use optical data as inputs, and Unimodal-NAS-SAR and Unimodal-NAS-Opt, which search for the optimal network architecture based on the unimodal characteristics instant of the handcrafted network architecture, whose classification results are present in rows #1–#4 of Table II; 2) the MDL methods: MultiCNN [21] aggregated the spectral and spatial features from different modalities based on the linear fusion strategy and then improved land cover classification, MBFNet [1], MLB [44], FBC [20], and LC-BCNet [21] have geared toward modeling bilinear to calculate mixing second-order moments to extract cross-modal high-order interaction of multimodal features, and CHGFNet [2] explored the efficiency of attention mechanisms in heterogeneous land cover classification, whose classification results are present in rows #3–#10 of Table II; 3) state-of-the-art multimodal NAS methods: Random search [36] and MFAS [51], whose classification results are present in rows #11–#12. The performance metrics, pixel accuracy (PA), mean pixel accuracy (MPA), mean intersection over union (MIoU), and frequency weighted intersection over union (FWIoU), are described below

$$PA = = \frac{\sum_{i=1}^{K} n_{(i,i)}}{\sum_{i=1}^{K} t_i} \tag{12}$$

$$\text{MPA} = \frac{1}{K} \sum_{i=1}^{K} \frac{n_{(i,i)}}{t_i} \tag{13}$$

$$\text{MIoU} = \frac{1}{K} \sum_{i=1}^{K} \frac{n_{(i,i)}}{t_i + n_{(j,i)} - n_{(i,i)}} \tag{14}$$

$$\text{FWIoU} = \frac{1}{\sum_{k=1}^{K} t_k} \sum_{i=1}^{K} \frac{t_i n_{(i,i)}}{t_i + n_{(j,i)} - n_{(i,i)}} \tag{15}$$

where $n_{(j,i)}$ is the pixels' number of class j that is misclassified to class i, $n_{(i,i)}$ is the pixels of class i that is properly classified, $t_i = \sum_{j=1}^{K} n_{(j,i)}$ denotes the pixel number of class i, and K is the number of land types.

We have gathered several experimental findings. First, our method provides the top heterogeneous land cover classification performance on all the datasets. Second, we use the DARTS algorithm to search for the optimal unimodal network architecture, Unimodal-NAS-Opt. and Unimodal-NAS-SAR, which only explores unimodal information for land cover classification; however, Unimodal-NAS-Opt. achieves the experimental results not inferior to MultiCNN with manual backbones, which means automatically network structure

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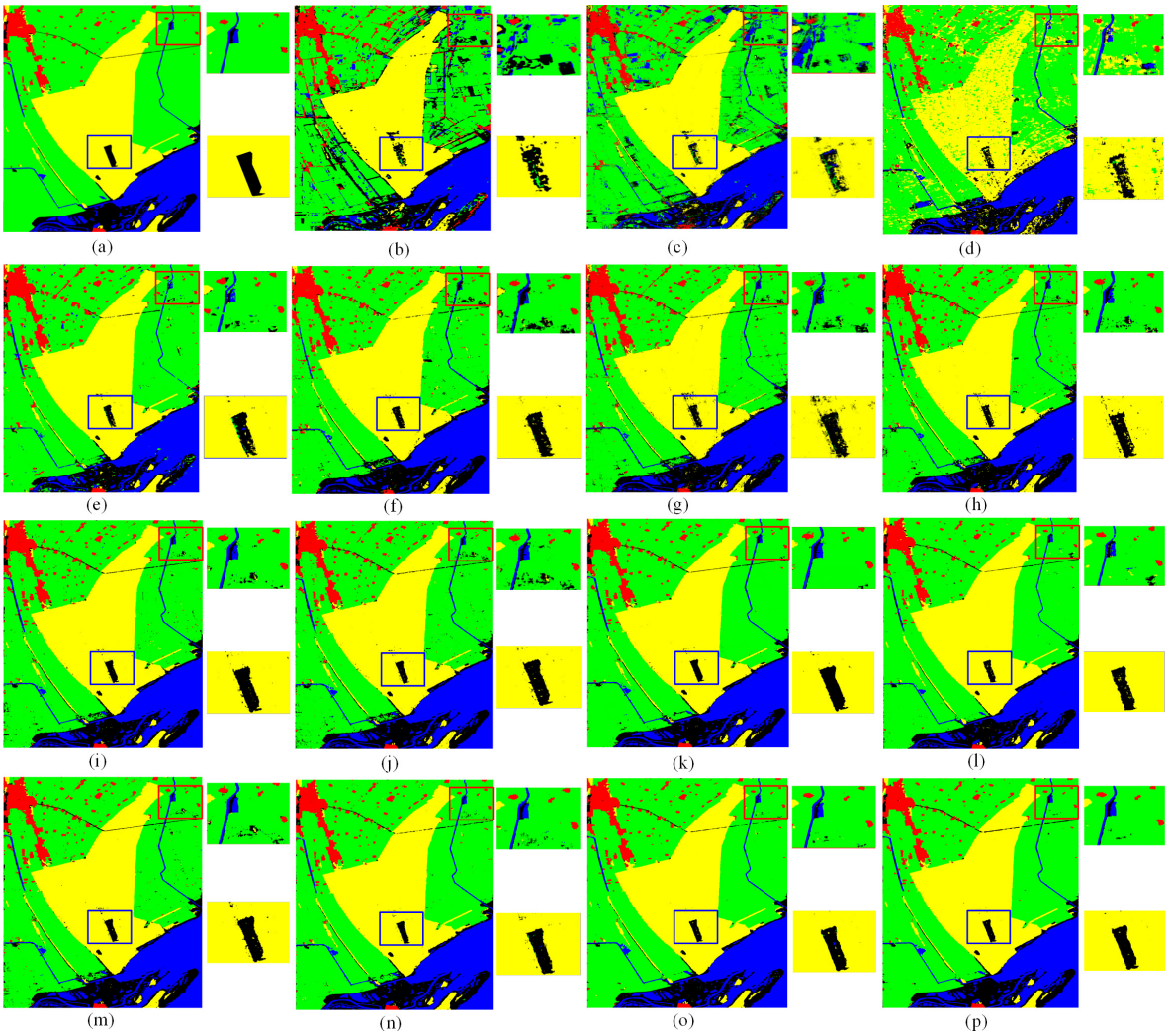


Fig. 10. Classification maps of different methods on PoDelta. (a) Ground truth. (b) SCNN [47]. (c) Suoyan *et al.* [49]. (d) Unimodal-NAS-SAR. (e) Unimodal-NAS-Opt. (f) CNN-Opt.&SAR. (g) MultiCNN [44]. (h) B-CNN [18]. (i) FBC [20]. (j) MLB [45]. (k) LC-BNet [21]. (l) CHGFNet [2]. (m) MBFNet [1]. (n) Random search. (o) MFAS [51]. (p) M^{2}SC-FAS (our).

design can rival or even surpass the human-invented architecture in terms of test set precisions. Third, exploring hierarchical optical and SAR feature fusion is in favor of land cover classification in comparison to our method and the second categories, mainly because the semantic consistency-based fusion architecture search strategy can search optimal hierarchical form between optical and SAR features. Finally, it is seen that the introduced semantic consistency constraint is beneficial for the learning of optimal multimodal dense fusion architecture in comparison to the fusion architecture search method. In a word, the NAS-based methods are effective and promising to automatically search suitable network architecture according to specific semantic features of different modalities.

Visually, Figs 8–10 show the classification score maps of the proposed M^{2}SC-FAS and contrast methods on the Libourne, Bergerac and PoDelta datasets, respectively. The proposed M^{2}SC-FAS has more coincident with the ground truth in comparison to other methods. To more intuitively the compare classification maps of different methods, we focus on some detail with the red and blue regions in Figs 8–10, respectively. In the red region of Fig. 8, many of the farmland and tree areas are not smooth, which were misclassified as resident area and industrial land by the comparison methods, whereas the farmland and tree areas are accurately identified by M^{2}SC-FAS. Similarly, in the blue regions, M^{2}SC-FAS provides smoothest visual results in comparison to other methods. Some details are shown in the red and blue regions of Fig. 9. In the red region, the proposed M^{2}SC-FAS provides the smoothest visual results in comparison to other methods; the farmland and tree areas are easily confused for the comparison methods. Similarly, in the blue region, the bare soil is misclassified as farmland by the comparison methods, whereas the bare soil areas are accurately identified by M^{2}SC-FAS. Similarly, in the detail regions of Fig. 10, the proposed M^{2}SC-FAS method provides more accurate boundaries and has the smoothest visual result in comparison to other methods.

F. Ablation Experiments

1) Validation of Modality-Specific Architecture Search: Recent studies [23], [37] revealed that pretraining methods oriented to natural/everyday images may not be the optimal solution to the tasks of RS image interpretation. Therefore,

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TABLE II

PA, MPA, MIoU, AND FWIoU OF DIFFERENT METHODS ON THE LIBOURNE, BERGERAC, AND PADELTA DATASETS

#	Methods	Libourne				Bergerac				PoDelta			
		PA	MPA	MIoU	FWIoU	PA	MPA	MIoU	FWIoU	PA	MPA	MIoU	FWIoU
1	SCNN (opt.) [47]	77.07	81.35	63.01	65.99	74.40	80.67	67.82	62.62	84.75	85.52	64.15	77.23
2	Suoyan *et.al* (opt.) [49]	80.70	84.99	69.24	69.94	80.93	85.84	75.50	68.99	89.08	87.51	70.26	82.17
3	Unimodal-NAS-SAR	71.42	79.01	53.53	58.09	66.93	75.39	58.74	51.41	86.93	89.66	73.22	77.91
4	Unimodal-NAS-Opt	84.82	90.63	72.23	75.24	87.01	91.89	84.35	77.87	95.73	94.79	84.36	92.34
5	CNN-Opt.&SAR	87.21	91.37	75.47	77.96	87.49	91.73	85.59	77.59	95.36	95.58	85.95	92.09
6	MultiCNN [44]	85.01	90.25	72.24	75.19	85.54	90.58	83.64	75.43	94.84	94.62	81.77	91.16
7	B-CNN [18]	87.78	92.47	76.85	79.25	88.81	92.54	88.55	80.71	95.98	96.03	85.71	92.86
8	FBC [20]	88.37	93.51	78.62	80.04	89.10	92.97	89.90	80.05	96.64	97.21	87.15	93.81
9	MLB [45]	88.23	92.64	78.05	79.94	88.83	92.83	87.51	79.54	96.27	97.09	86.71	93.34
10	LC-BCNet [21]	89.19	94.09	79.58	81.29	89.72	93.23	88.45	81.06	97.14	97.82	89.27	94.77
11	CHGFNet [2]	89.01	94.23	79.27	80.89	89.56	93.41	89.29	80.82	97.86	97.11	88.59	93.54
12	MBFNet [1]	88.31	92.79	77.45	78.92	88.39	92.43	88.92	79.23	96.18	96.53	84.12	92.03
13	Random search	88.73	93.86	79.55	80.72	89.07	92.46	89.34	82.71	97.09	96.99	91.43	94.55
14	MFAS [51]	89.64	93.54	79.28	82.61	89.61	92.97	89.46	83.04	97.73	97.45	92.03	95.32
15	M^2SC-FAS (Our)	**91.03**	**94.90**	**80.36**	**83.05**	**90.68**	**93.74**	**90.54**	**83.49**	**98.94**	**98.57**	**93.01**	**96.44**

TABLE III

LAND COVER CLASSIFICATION PRECISIONS FOR UNIMODAL METHODS WITH DIFFERENT BACKBONES ON LIBOURNE, BERGERAC, AND PODELTA

Methods	Backbones	Libourne					Bergerac					PoDelta				
		Para.	PA	MPA	MIoU	FWIou	Para.	PA	MPA	MIoU	FWIou	Para.	PA	MPA	MIoU	FWIou
CNN-SAR	AlexNet	4.93M	68.61	72.29	49.58	53.98	4.98M	62.01	73.60	51.34	45.77	3.94M	81.90	87.10	62.59	70.85
CNN-SAR	VGG16	14.97M	69.46	78.57	50.66	55.07	14.98M	65.91	74.82	55.91	50.05	14.78M	84.11	89.13	69.04	73.77
CNN-SAR	ResNet34	63.22M	70.10	77.74	53.41	57.16	63.23M	63.36	72.66	51.97	47.55	63.02M	85.21	88.66	69.33	75.39
CNN-SAR	ResNet101	83.42M	70.97	78.69	53.27	57.83	83.43M	63.74	73.47	52.12	47.99	83.22M	85.90	**89.81**	70.59	76.37
Unimodal-NAS-SAR	–	14.12M	**71.42**	**79.01**	**53.53**	**58.09**	8.82M	**66.93**	**75.39**	**58.74**	**51.41**	17.01M	**86.93**	89.66	**73.22**	**77.91**
CNN-Opt.	AlexNet	4.93M	80.75	86.08	67.09	68.01	4.98M	84.65	89.20	81.23	73.99	3.94M	92.21	92.07	82.24	86.24
CNN-Opt.	VGG16	14.97M	83.20	90.24	70.62	72.88	14.98M	85.93	91.36	83.48	75.87	14.78M	93.81	93.82	82.84	89.35
CNN-Opt.	ResNet34	63.22M	84.01	90.05	71.86	73.51	63.23M	86.34	91.27	82.64	75.94	63.02M	94.42	94.70	83.37	90.38
CNN-Opt.	ResNet101	83.42M	84.43	90.53	73.03	74.48	83.43M	86.52	90.07	**84.69**	77.44	83.22M	95.23	**95.45**	83.49	91.87
Unimodal-NAS-Opt.	–	6.91M	**84.92**	**90.93**	**73.23**	**75.24**	8.51M	**87.01**	**91.89**	84.35	**77.87**	5.02M	**95.73**	94.78	**84.36**	**92.32**

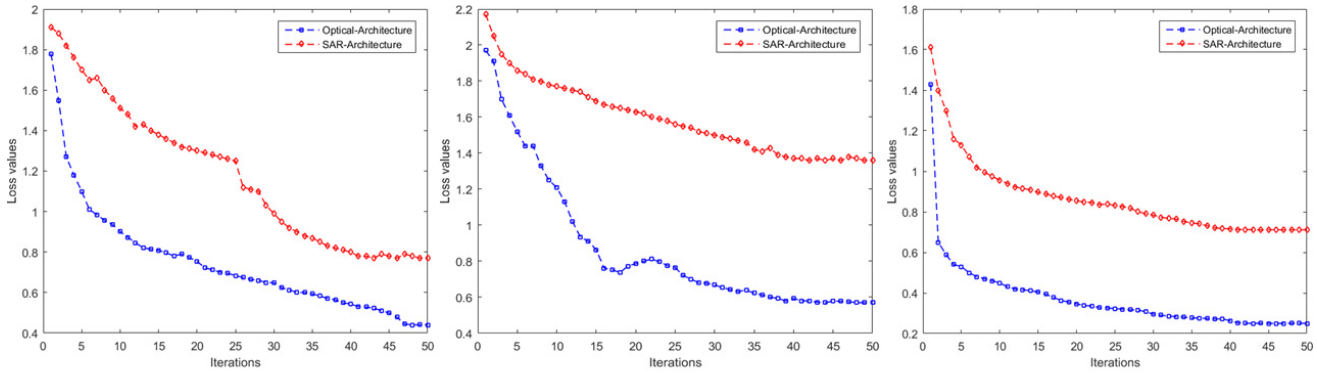


Fig. 11. Loss curves regarding architecture optimization for the optical and SAR modalities on (a) the Libourne, (b) Bergerac, and (c) PoDelta datasets, respectively.

we automatically design the optical- and SAR-specific architectures based on the architecture search algorithm for land cover classification. Table III shows comparison of the M^2SC-FAS and contrast methods based on different pretrained backbones. As expected, the searched modality-specific architectures consistently achieve significant improvement compared with the pretrained models, which indicates that the searched modality-specific architectures has powerful expressive and discriminative ability. In addition, we show the number of parameters in "Para." column. Note that the Unimodal-NAS-SAR and Unimodal-NAS-Opt. require less parameters, apart from CNN-SAR and CNN-Opt. with AlexNet, and Unimodal-NAS-SAR and Unimodal-NAS-Opt. achieve classification results better than the comparison methods.

Furthermore, we present loss curves regarding architecture optimization for the optical and SAR modalities on the Libourne, Bergerac, and PoDelta datasets, respectively. It is seen that the loss value gradually decreases as the number of iterations increases in Fig. 11. This means that the proposed M^2SC-FAS can learn optical- and SAR-specific feature extractors with the help of the network architecture search algorithm, efficiently.

2) Validation of M^2SC-FAS : MSC constrain term guides the learning of multimodal dense fusion architecture search space $\mathcal{B}$ and then obtain the optimal way of mix features with varying

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TABLE IV
LAND COVER CLASSIFICATION PRECISION FOR METHODS WITH AND WITHOUT MSC

#	Methods	MSC	PA	MPA	MIoU	FWIoU	Datasets
1	M^{2}SC-FAS	×	89.78	94.01	80.01	82.33	Libourne
2	M^{2}SC-FAS	✓	**91.03**	**94.91**	**80.35**	**83.02**	
3	M^{2}SC-FAS	×	90.08	93.22	90.09	83.12	Bergerac
4	M^{2}SC-FAS	✓	**90.69**	**93.74**	**90.56**	**83.47**	
5	M^{2}SC-FAS	×	97.45	97.67	92.67	96.01	PoDelta
6	M^{2}SC-FAS	✓	**98.94**	**98.59**	**92.99**	**96.44**	

levels at different modalities. To explore the effectiveness of MSC constrain term, we present the classification accuracies of M^{2}SC-FAS with and without MSC constrain term in Table IV. We assemble several experiment findings. First, M^{2}SC-FAS with the MSC constraint term achieves better classification accuracies. Second, MSC constraint term can explore the optical and SAR features with the same semantics to achieve efficient fusion, which is in favor of MLCC. In short, MSC searches automatically the multimodal dense fusion architecture, instant of artificially designed.

IV. CONCLUSION

This article proposes a novel multimodal architecture search framework, namely, M^{2}SC-FAS in continuous search space for land cover classification using optical and SAR RS images. Unlike some previous MDL approaches that used Siamese and pseudo Siamese network with handcrafted backbone as feature extractor, the proposed M^{2}SC-FAS first searches the optimal optical-specific and SAR-specific architectures to extract hierarchical optical and SAR features. Then, the semantic consistency constraint based on RDM is used to guide the search of multimodal dense fusion architectures, which find the optimal way of mix features with varying levels in different modalities to obtain their complementary performance on land properties. Finally, the basis of the curriculum learning strategy is adopted on multimodal dense fusion architecture search, allowing each branch to be initialized with existing weights of modality-specific architectures. Extensive experiments show superior performances of our method on three representative broad coregistered optical and SAR datasets.

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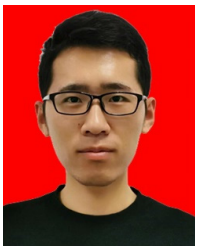
Xiao Li (Student Member, IEEE) received the M.S. degree in control science and engineering from Xiangtan University, Xiangtan, China, in 2018. He is currently pursuing the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China.

His research interests include image processing and pattern recognition, sparse representation, deep learning, and remote sensing image applications.



Lin Lei received the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2008.

She is currently an Associate Professor with the School of Electronic Science, National University of Defense Technology. Her research interests include computer vision, remote sensing image interpretation, and data fusion.



Caiguang Zhang received the M.S. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2019, where he is currently pursuing the Ph.D. degree with the State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System.



Gangyao Kuang (Senior Member, IEEE) received B.S. and M.S. degrees in geophysics from the Central South University of Technology, Changsha, China, in 1988 and 1991, respectively, and the Ph.D. degree in communication and information from the National University of Defense Technology, Changsha, in 1995.

He is currently a Professor with the School of Electronic Science, National University of Defense Technology. His research interests include remote sensing, SAR image processing, change detection, SAR ground moving target indication, and classification with polarimetric SAR images.

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